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Q. Geometric Mean, G.M = 29.09.2026

• Ungroup:

$$(x_1 \cdot x_2 \cdot x_3 \dots x_n)^{1/n}$$

• here, n is total number of observation.

$$\rightarrow \left(\prod_{i=1}^n x_i \right)^{1/n}$$

• Group:

$$G.M = \left(\prod_{i=1}^n f_i x_i \right)^{1/n}$$

• x_i = observations

• f_i = frequency

Harmonic Mean, H.M. =

• Ungroup:
$$\frac{n}{\sum_{i=1}^n \frac{1}{x_i}}$$

• Group:
$$\frac{n}{\sum_{i=1}^n \frac{f_i}{x_i}}$$

Find the relation between A.M, G.M and H.M / Show that $A.M > G.M > H.M$.

Proof:

Let, x_1 and x_2 be two observations and they are positive integers.

Then, $A.M = \frac{x_1 + x_2}{2}$

$G.M = (x_1 x_2)^{1/2} = \sqrt{x_1 x_2}$

Since,

x_1 and x_2 are non-negative integers, x_1 and x_2 both are real.

So, $\sqrt{x_1}$ and $\sqrt{x_2}$ are also real numbers.

We know,

Square of two real numbers never be negative. or, $(\sqrt{x_1} - \sqrt{x_2})^2 \geq 0$

$$\text{i.e. } (\sqrt{x_1} - \sqrt{x_2})^2$$

$$= x_1 - 2\sqrt{x_1 x_2} + x_2$$

$$\Rightarrow x_1 + x_2 - 2\sqrt{x_1 x_2} \geq 0$$

$$\Rightarrow x_1 + x_2 \geq 2\sqrt{x_1 x_2}$$

$$\Rightarrow \frac{x_1 + x_2}{2} \geq \sqrt{x_1 x_2}$$

$$\Rightarrow A.M \geq H.M \quad \text{--- (i)}$$

$$H.M = \frac{2}{\frac{1}{x_1} + \frac{1}{x_2}} = \frac{2x_1 x_2}{x_1 + x_2}$$

Since, x_1 and x_2 are real,

$\frac{1}{\sqrt{x_1}}$ and $\frac{1}{\sqrt{x_2}}$ are also real.

and we know square of two real numbers

never be negative

$$\text{i.e. } \left(\frac{1}{\sqrt{x_1}} - \frac{1}{\sqrt{x_2}} \right)^2 \geq 0$$

$$\Rightarrow \frac{1}{x_1} - \frac{2}{\sqrt{x_1 x_2}} + \frac{1}{x_2} \geq 0$$

$$\Rightarrow \frac{x_1 + x_2}{x_1 x_2} \geq \frac{2}{\sqrt{x_1 x_2}}$$

$$\Rightarrow \sqrt{x_1 x_2} \geq \frac{2 x_1 x_2}{x_1 + x_2}$$

$$\Rightarrow G.M \geq H.M$$

From eqn (i) & (ii) we have

$$A.M \geq G.M \geq H.M$$

Hence, $A.M. > G.M > H.M$ (showed)